Preprint, not peer reviewed

Liabut: a pre-registered computational test of how Timor's Austronesian languages branch

Dennis Crowley

Phaistos Labs

Abstract

Two classifications of Timor's Austronesian languages disagree on where Lakalei-Idate belongs and on whether Welaun is a Central Timor language or a dialect of Tetun. Hull groups Lakalei-Idate with Mambae, Kemak and Tokodede as Ramelaic; Grimes and Edwards (2026) place it in Timor-Babar, apart from Central Timor. This paper tests both against basic vocabulary from LexiRumah, under a protocol frozen before analysis and nine dated addenda. The pre-registered bootstrap rule leaves both questions unresolved. Bayesian analyses of 24 lects support Central Timor with Welaun (posterior 0.86, ln BF +10.3) and place Lakalei-Idate as its sister (0.91). No analysis gives lexical support to Hull's Fabronic, to Timor-Babar or to Eastern Timor, a result that survives consensus cognate coding, a retention classifier validated against experts, and word lists collected independently of Grimes and Edwards; Timor-Babar, defined by sound changes, is not refuted by this. On those independent lists Lakalei-Idate groups with Mambae, Kemak and Tokodede at 97%. The arguments for its eastern placement do not discriminate once tested: the coded sound changes cost the same on either tree, and a borrowing simulation cannot separate the placements when borrowing runs both ways. Contact signals other than Welaun's Tetun loans do not survive calibration or multiple-testing correction, and the confirmatory predictions failed against Tetun. Lakalei-Idate's position remains formally unresolved but leans towards Central Timor, as an old member of a central Timor dialect chain. All data and code are public.

Keywords: Timor; Austronesian; subgrouping; computational phylogenetics; language contact; pre-registration

1 Introduction

Two classifications of the Austronesian languages of Timor disagree. Hull (1998, 2001) groups them all as *Timoric*, split into *Ramelaic* (Mambae, Kemak, Tokodede and Idalaka, that is Idaté and Lakalei) and *Fabronic* (Tetun, Habun, the Kawaimina languages, Galoli, Wetarese, Dawan and Bekais or Welaun). He proposed that Ramelaic descends from Old Idalaka (as summarised by Albuquerque 2009). Carvalho (2001, cited in Fogaça 2015) proposed the same two-way split.

Table 1. Data sources.

Source	Contents	Use
LexiRumah 3.0.1 (Kaiping, Edwards and Klamer 2019)	70 Timor-region lects, 152 concepts	Main lexical data (D1)
ABVD (Greenhill, Blust and Gray 2008)	30 lects with expert cognate judgements; Proto-Malayo-Polynesian (PMP) forms	Validation (D2); PMP etyma
Grimes and Edwards (2026), LaTeX source	257 comparative tables, 14,401 forms, reflexes marked by the authors	Sound-change atlas and characters
ASJP v21 (Wichmann, Holman and Brown)	40-item lists, including Idaté from Purwa et al. (1994)	Independent check
Alcantara (2015)	184 Idaté forms from fieldwork in Laclubar	Independent check
Gray et al. (2009) via phlorest	998 posterior trees	Comparison

Edwards (2019, 2021) and Grimes and Edwards (2026) instead make *Central Timor* (Mambae, Kemak, Tokodede and Welaun) a separate branch of Malayo-Polynesian, defined by *ŋ > *g > k, and place every other Austronesian language of Timor, Idaté and Lakalei included, in *Timor-Babar*, defined by *p > *h. Grimes and Edwards do not cite Hull.

The two disagree most sharply on two points. **C1**: does Lakalei-Idate belong with Central Timor, as in Hull, or with Timor-Babar, as in Edwards and Grimes and Edwards? **C2**: is Welaun a dialect of Tetun, as Hull (1998) first had it, or a Central Timor language (Edwards 2019)?

Grimes and Edwards argue from historical phonology, supported by morphosyntax and the lexicon. The only earlier phylogeny covering Timor, Gray, Drummond and Greenhill (2009), includes Tetum, Mambai and Kemak but none of Idaté, Lakalei, Welaun, Habu, Midiki or Helong. This paper asks what the basic vocabulary says, using a pre-registered design in the spirit of Robinson and Holton (2012), who used lexical phylogenetics where sound changes had left the subgrouping of Alor-Pantar unresolved. It then asks how far the lexical signal reflects inheritance rather than contact, and how robust each answer is.

2 Data

Each lect needed at least 100 of the 152 concepts. Forms that look like loans from Indonesian or Portuguese, forms that look like Tetun, and contact sets shared with neighbouring Papuan languages were flagged and screened in sensitivity variants. Idaté, Lakalei, Welaun, Habu, Midiki and Helong are absent from ABVD; this is their first cognate-based phylogenetic placement. Idaté, Welaun, Habu and Helong do have short ASJP lists. The Idaté and Lakalei lists come from surveys by Klamer (2002) and Dawson (2014), not from Edwards or Grimes. Figure 1 shows the 24 lects used in the model-based analyses, one per language.

3 Methods

Pre-registration. The protocol was frozen and fingerprinted (SHA-256) before any analysis. Nine dated addenda, each pushed to a public repository before the tests it governs, extended it.

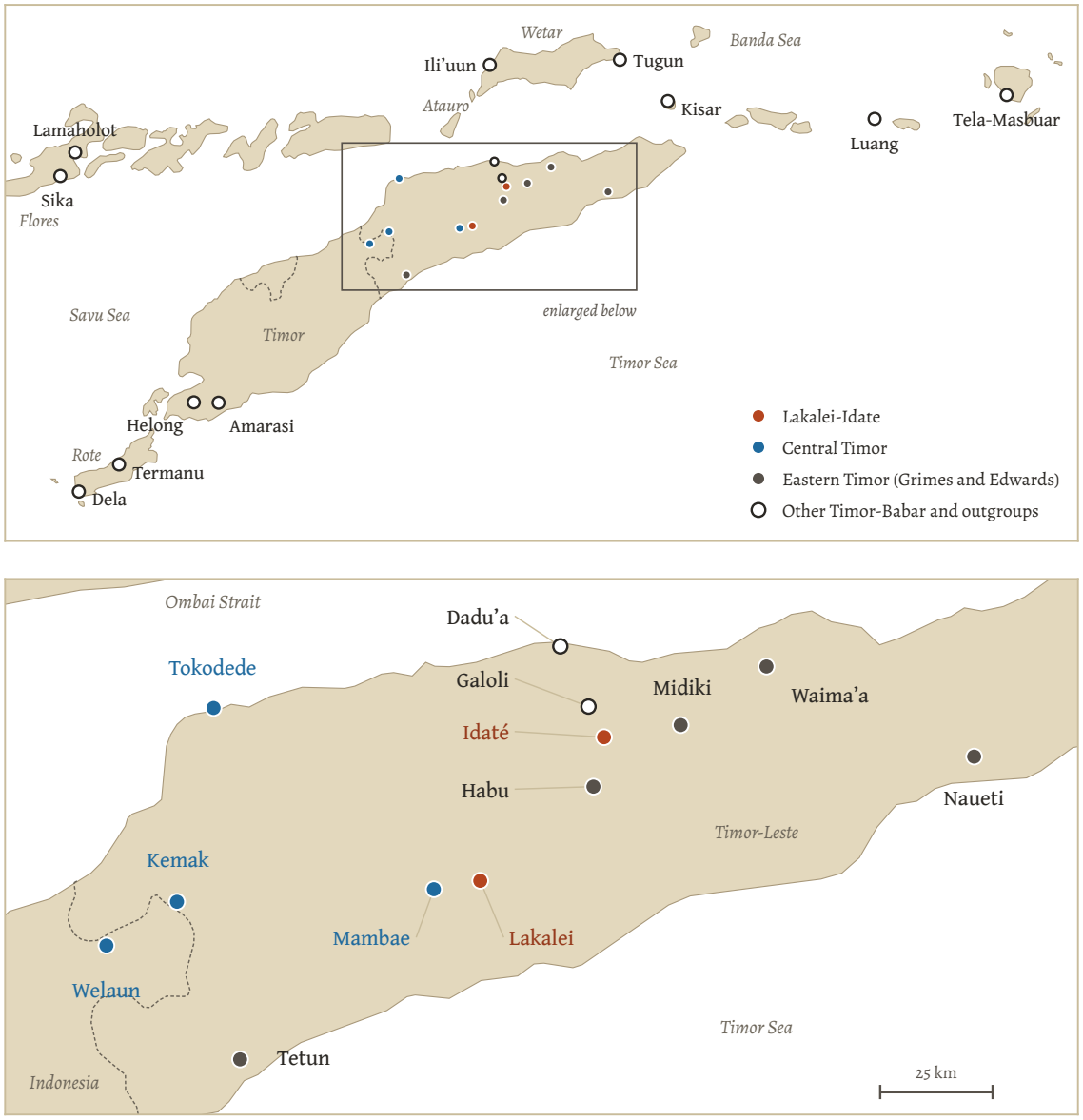


Figure 1. The 24 lects used in the Bayesian and maximum-likelihood analyses, one per language. Lakalei-Idate (red) lies between Central Timor (blue) and the Eastern Timor languages of Grimes and Edwards (dark grey); the dashed line is the border between Timor-Leste and Indonesia. Coordinates from LexiRumah; coastlines from Natural Earth.

Everything after the frozen protocol is post hoc and is labelled as such, except two replications in Addendum 9 whose predictions were stated in advance. The pre-registered decision rule required a contested clade to reach 70% bootstrap support in both UPGMA and NJ trees, on the primary data and on two screened variants.

Cognates. Automatic cognate detection used LexStat (threshold 0.55) and SCA (0.45) with Infomap clustering in LingPy (List, Greenhill and Gray 2017). Scored against the ABVD expert judgements, LexStat reaches a B-cubed F of 0.81 (precision 0.88, recall 0.75); distances from the automatic cognates match ABVD expert distances at Mantel r = 0.90. A gate of known low-level groups had to be recovered before any contested clade was scored.

Trees. Distance trees (UPGMA, NJ) with 1,000 concept bootstraps were built for 30 hypothesis clades. Treelikeness was measured with delta scores and Q-residuals. Bayesian

analyses in MrBayes 3.2.7a used 24 lects and 1,353 binary cognate characters (restriction model with gamma rate variation). Each contested clade was run forced and forbidden, and Bayes factors were computed from stepping-stone marginal likelihoods. Maximum-likelihood trees used IQ-TREE 2.3.6 with ultrafast bootstrap.

Sound changes. The marked reflexes in Grimes and Edwards's tables were turned into an expert reflex atlas; LexiRumah forms aligned to ABVD PMP forms gave an automatic atlas. The two agree on 94% of tokens (249 of 264). Eight diagnostic changes, later eleven, were coded as binary characters and combined with the lexicon.

Inheritance and contact. Each cognate set was classed as a PMP retention or an innovation (the PMP word was replaced). The rule was validated against ABVD's experts and improved (Section 7). On that basis:

1. Exclusive shared innovations were counted, with a control for geographic distance.
2. A triplet site-pattern test, the lexical counterpart of the ABBA-BABA statistic, counts for three groups the concepts on which exactly two share an innovation. The largest count marks the tree pair, and an imbalance between the other two marks contact. It was run separately on the 60 Leipzig-Jakarta concepts (Tadmor 2009) and on the rest, with Welaun's known Tetun contact as a check, and was later calibrated by simulation.
3. A split test asks which member of Lakalei-Idate carries each signal.
4. Approximate Bayesian computation (ABC) estimates the borrowing that each placement of Lakalei-Idate would need.
5. Historical glottometry (Kalyan and François 2018) and strict-clock analyses give a non-tree and a dated view.

Robustness (Addendum 9). Ten further analyses target the limitations of the first version: multiple-testing correction (Holm 1979; Benjamini and Hochberg 1995), replication with other dialect representatives and with lists independent of Grimes and Edwards, a consensus of five cognate codings, a retention classifier validated against experts, the sound changes scored by parsimony on the rival trees, a simulated null and power analysis for the triplet test, an ABC with borrowing in both directions and random-forest model choice (Pudlo et al. 2016), a correspondence-based loan test, and a clock rooted on the outgroup. Analyses are numbered E1 to E34 as in the repository.

4 Results: the pre-registered tests

The gate passed under an addendum adopted before any contested clade was scored. Only low-level groups reach the 70% bar in both distance trees: Lakalei-Idate, Kawaimina, Wetar-Atauro, Nuclear Rote, Kisar-Luang and Babar. Hull's Fabronic, Timor-Babar and Eastern Timor score 0 to 3%. Ramelaic reaches 88 to 95% only when Tetun look-alikes are removed, a variant the protocol flagged in advance as confounded, because removing those words removes the evidence Edwards's placement predicts. Mean delta is 0.24 for 37 lects and 0.33 for one lect per language; the most reticulate lects are the contested ones (Welaun, Tokodede, Galoli, Lakalei, Idaté, Habu).

The pre-registered decision was **unresolved** for both C1 and C2, because neither contested clade reached the bootstrap bar. The Bayesian analyses, which the rule did not use for the decision, favour Central Timor with Welaun, with Lakalei-Idate as its sister (Figure 2).

Table 2. Bayesian constraint tests and posterior clade probabilities. Stepping-stone runs agree within 0.8 log units; the unconstrained runs converged (split-frequency difference 0.005, minimum ESS 1,074).

Bayesian test (24 lects)	ln BF, forced against forbidden	Unconstrained posterior
C1: Hull's Ramelaic	+8.0	0.13 as drawn (without Welaun)
C2: Central Timor with Welaun	+10.3	0.86
Lakalei-Idate + Central Timor with Welaun		0.91
Timor-Babar	−7.0	0.00
Eastern Timor, with or without Lakalei-Idate		0.00
Welaun + Tetun		0.00

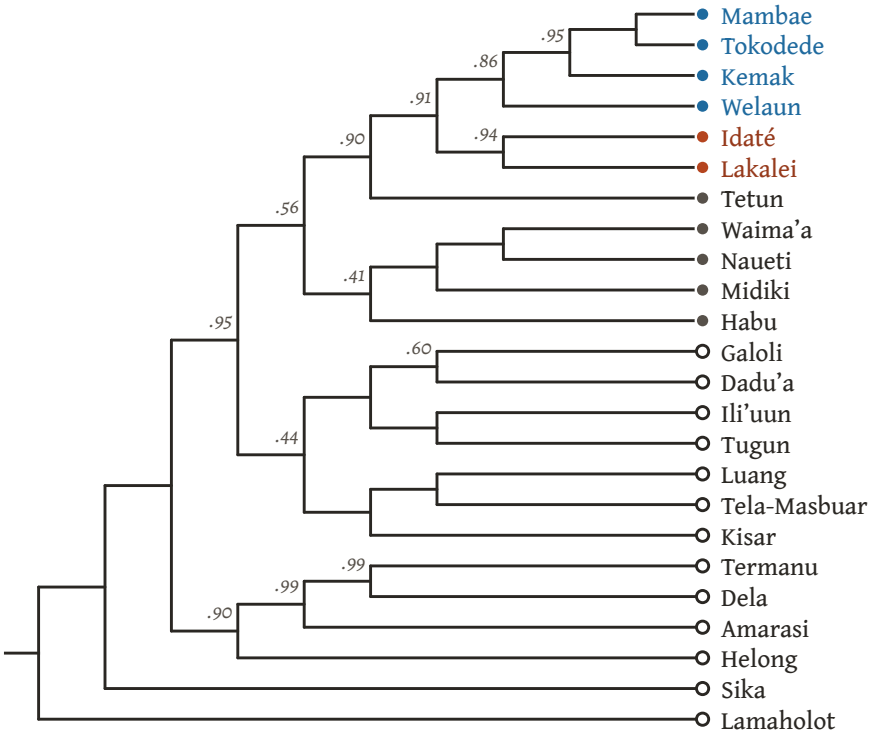


Figure 2. Consensus tree (all compatible splits) of the unconstrained Bayesian analysis, 24 lects and 1,353 cognate characters, drawn as a cladogram and rooted on Lamaholot. Numbers are posterior probabilities; unlabelled nodes exceed 0.99.

5 Results: sound changes and combined evidence

Lakalei-Idate's sound changes resemble Timor-Babar's: it has *p lenition and *ŋ > n, and lacks Central Timor's *ŋ > g/k and *j > l (Figure 3). Two facts weaken these as diagnostics. Welaun, a Central Timor language on every other count, also has *p > h; and *ŋ > n also occurs in Sika, a Flores outgroup. Hull's own Ramelaic evidence, initial *b- > h, is visible in Grimes and Edwards's tables: Lakalei-Idate shares it with Mambae, Kemak and Tokodede, against Tetun's *b- > f and Kawaimina's *b- > w, but Galoli and Ili'uun share it too.

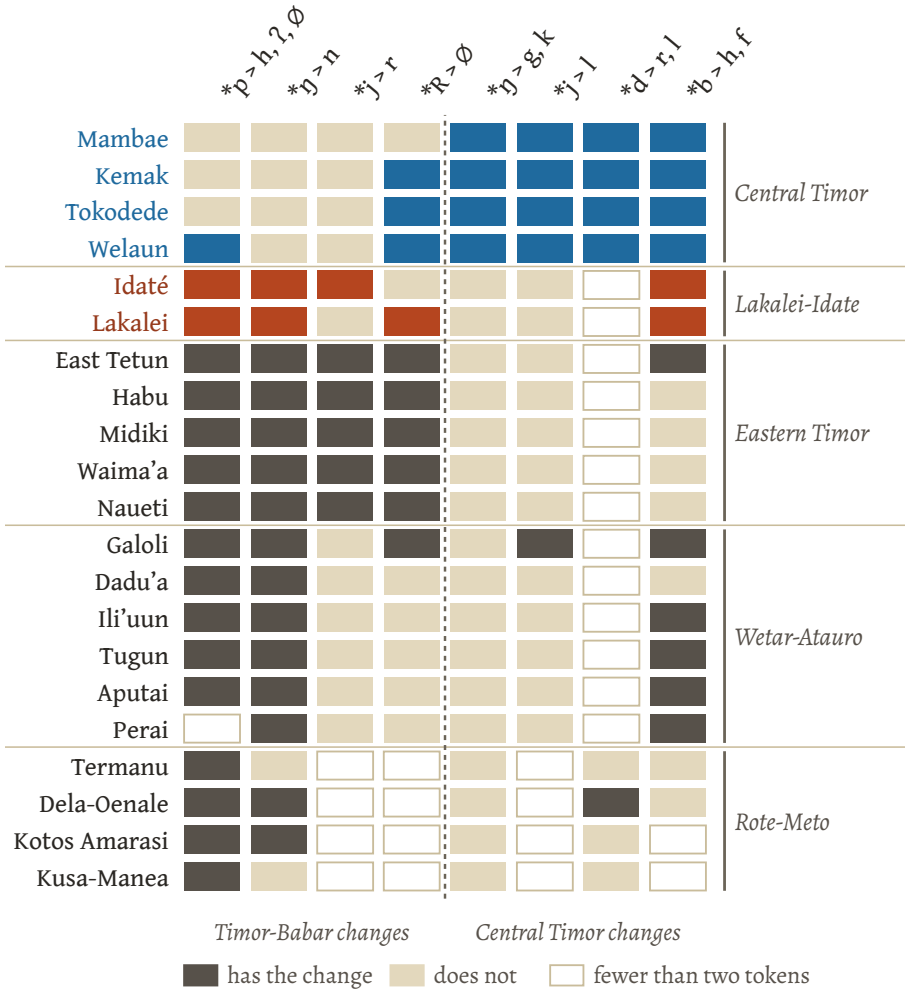


Figure 3. Eight diagnostic sound changes coded from the reflexes Grimes and Edwards (2026) mark in their comparative tables, for the languages of Timor, Wetar-Atauro and Rote-Meto (Southwest Maluku is omitted here and shown in the repository). Outlined cells rest on fewer than two expert tokens. The last column merges *b > h and *b > f, so it is not the same character as Hull's initial *b- > h.

The coded changes do not, in fact, favour Grimes and Edwards's tree. Scored by Fitch parsimony, the eleven characters need 30 steps on the lexical ML tree (Lakalei-Idate beside Central Timor), 30 on the tree constrained to Lakalei-Idate + Eastern Timor, and 29 on a tree constrained to Grimes and Edwards's top split, which costs 12 log-likelihood units in the lexicon. Because Tetun is sister to Lakalei-Idate + Central Timor in the lexical tree, the eastern-looking changes can sit below that group at no extra cost. Pooling expert, automatic and Alcantara tokens also shows how thin some cells are: Idaté's *j > r rests on 5 of 12 tokens,

Table 3. Combined analyses.

Analysis (24 lects)	Lakalei-Idate + Central Timor	Timor-Babar
Vocabulary only, ML (UFBoot %)	65	0
Vocabulary + 17 sound-change characters, ML	56	0
Vocabulary + 8 diagnostic changes, ML	80	0
Vocabulary + 11 changes (with Hull's *b, *k), ML	75	0
Vocabulary + 17 characters, Bayesian posterior	0.88	0.00
Vocabulary without 21 probable Tetun loans, ML	57	0

and both members have *d > r, l, a Central Timor change, in 8 of 9 and 5 of 7 tokens (mostly automatic).

Support depends on how cognates are coded, a check modelled on Kaiping and Klamer (2022). Lakalei-Idate + Central Timor gets 92% with SCA cognates and 65% with the primary coding, but 15 to 20% at LexStat thresholds of 0.35 and 0.75; at those thresholds Lakalei-Idate still groups with Mambae, Kemak and Tokodede, alone or with Tetun and Habu added. LexiRumah's own published cognates put it with neighbouring Habu (51%); that coding fits the expert judgements worst (Mantel r 0.70) and does not recover Central Timor. Timor-Babar receives no support under any of the five codings.

6 Results: inheritance or contact for Lakalei-Idate

Shared innovations. Lakalei-Idate shares 7 exclusive innovations with Central Timor (among them 'cry' *sɛro*, against *tanis* < *taŋis almost everywhere else) and 9 with the Eastern Timor groups combined.

Triplet test. On stable concepts, Lakalei-Idate + Nuclear Central Timor (Mambae, Kemak, Tokodede) is the largest pair against every other group tested with the representative lists (Figure 4), though not against Tetun Dili (Section 7). The test recovers Welaun's Tetun contact, and places it where contact should be: in borrowable vocabulary (Z = 6.9), not in stable vocabulary (Z = 0.6). The first version also read Lakalei-Idate's extra sharing with Tetun, Habu and Kawaimina as possible contact. Calibration withdraws that reading. With the tree's unequal branch lengths and no borrowing at all, simulation gives D values centred on 0.44 to 0.47 for these triplets, and every observed value (0.12 to 0.65) lies inside that null; the jackknife rule (Z ≥ 2) fires in 27% of contact-free simulations. Welaun's borrowable-word Z exceeds 99.8% of the null. Once calibrated, the test has a 5% false-positive rate but detects 20% borrowing from Habu only 30% of the time.

Which half carries the signal. In stable vocabulary, five of the eight Central Timor items are shared by both Idaté and Lakalei, so they predate the split between them. In borrowable vocabulary, six of eight are found only in Lakalei and match Mambae, spoken 10 km away ('cloud' *sabai*, 'neck', 'eat' *mia* ~ *mua*). The counts are small (binomial p = 0.11), and a correspondence test finds these items no closer to identity than inherited words, so the recent-loan reading is not confirmed.

How much borrowing each placement needs.

The first simulation preferred Grimes and Edwards's placement, but only with each model

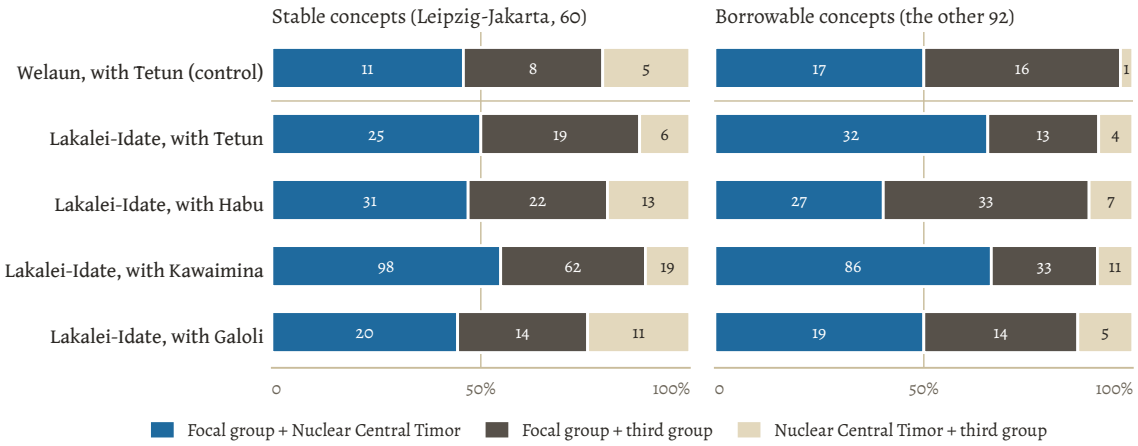


Figure 4. Triplet counts on stable (Leipzig-Jakarta) and borrowable concepts. Each bar is one triplet of the focal group (Welaun or Lakalei-Idate), Nuclear Central Timor (Mambae, Kemak and Tokodede) and a third group; the numbers count concepts on which exactly that pair shares an innovation. Welaun, whose Tetun contact is documented, is the control. Against a simulated null with unequal rates, only Welaun's excess in borrowable words stands out.

Table 4. Approximate Bayesian computation. The one-way models allow borrowing from the other side only; the two-way models allow it from both, with 36 summary statistics. Random-forest prior error for the two-way models: 36%.

ABC, 20,000 runs per model	Sister to Central Timor	Within Eastern Timor
One-way borrowing (E21): posterior	0.125	0.875
One-way: borrowing needed, stable, median (90%)	16% (1–41%) from the east	28% (4–53%) from Central Timor
Two-way borrowing (E32): random-forest votes	0.45	0.55
Two-way: rejection posterior	0.42	0.58
Two-way: Central Timor borrowing, stable	24% (4–52%)	25% (4–55%)
Two-way: posterior predictive checks failed (of 36)	0	0

borrowing from one side. Allowing borrowing in both directions removes the preference: neither placement reaches the 0.9 bar, both reproduce every summary statistic, and the random forest misclassifies 36% of its own simulations. The summary counts cannot separate descent from borrowing here. On the full lexical likelihood, the unconstrained tree is better than the constrained one by 10.5 log units.

Glottometry and dating. Glottometry finds cohesive groups only at low levels. Lakalei-Idate is cohesive on its own (subgroupiness 1.42) and tied almost equally weakly to Central Timor (0.06) and to Eastern Timor (0.08); Hull's Ramelaic scores 0. Under a strict clock rooted on the Flores outgroup, about 77% of posterior samples place Lakalei-Idate with Central Timor (Habu 11%), and their split (relative height 0.48) is barely older than Central Timor's own diversification (0.47); the earlier run, whose clock put the root on Tela-Masbuar, gave the same

Table 5. What Addendum 9 checked, and what came out.

Limitation of the first version	Check (Addendum 9)	Outcome
About twenty post hoc tests, uncorrected	Holm and Benjamini–Hochberg over 40 tests (E25)	Only Welaun's Tetun contact survives
Results could hinge on the chosen dialect lists	Every combination of 11 Mambae, 8 Kemak, 2 Idaté and 2 Tetun lists, predictions stated in advance (E26)	Stable-vocabulary pairing with Central Timor holds against Habu and Kawaimina (100%) but not Tetun (47%, failing on Tetun Dili)
LexiRumah shares sources and an editor with Grimes and Edwards	Lists collected by others only, predictions stated in advance (E28)	Lakalei-Idate + Nuclear Central Timor 97%; Timor-Babar and Lakalei-Idate + Eastern Timor 0%; not a same-collector effect; stable pairing fails narrowly against Tetun (45 to 44)
Automatic cognates; coding dependence	Consensus of five codings (E27); reviewer packet	Timor-Babar and Lakalei-Idate + Eastern Timor 0%; Lakalei-Idate + Nuclear Central Timor 55%
Retention classifier misclasses changed retentions	Validated on 3,801 expert-judged forms; reflex-aware rule (E30)	Balanced accuracy 0.755 → 0.791; main patterns unchanged
Triplet test assumes equal rates	Simulated null and power (E31)	Contact readings for Lakalei-Idate withdrawn; Welaun's stands
Borrowing simulation one-directional, 18 counts	Two-way ABC, 36 statistics, random forests, predictive checks (E32)	Placements indistinguishable
Sound changes taken as pointing east	Parsimony on rival trees; token intervals (E29)	30 steps on either placement
Comparative method not applied	Identity of aligned forms against inherited baseline (E33)	No loan signal; low power
Clock rooted on a long, sparse branch	Root constrained to the outgroup (E34)	Same picture: 77% Central Timor side

picture (80%; 0.29 against 0.27). A contact-aware phylogeny (contacTrees; Neureiter et al. 2022) did not reach a stable solution: it kept adding contact edges, each moving 14 to 23% of the vocabulary, so it is not reported further.

7 Robustness and replication

The two confirmatory replications give a mixed but informative result. Lists independent of Grimes and Edwards strengthen the tree placement of Lakalei-Idate beside Mambae, Kemak and Tokodede (97%), and in that tree Lakalei falls inside Nuclear Central Timor (94%) while Idaté-Cribas stays outside. Adding Klamer's other 2002 lists shows this is not a collector effect: his Tetun Dili list joins Tetun Suai (100%), not his Mambae and Kemak lists. The stable-vocabulary pairing replicates against Habu and Kawaimina in every combination and

Table 6. Triplet tests on other disputed placements. The tree pairs are robust to the classifier change; none of the contact excesses survives multiple-testing correction.

Triplet	Counts, all concepts (stable)	Reading
Habu, Tetun, Kawaimina	Habu+Tetun 15 (5); Habu+Kawaimina 11 (5); Tetun+Kawaimina 4 (2)	Unresolved; the Tetun link is in borrowable words
Waima'a, Naueti, Midiki	Waima'a+Naueti 10 (7); Waima'a+Midiki 7 (4); Naueti+Midiki 2 (1)	Leans towards Grimes and Edwards's Waima'a-Naueti
Galoli, Wetar, Dadu'a	Galoli+Dadu'a 20 (13); Galoli+Wetar 6 (2); Wetar+Dadu'a 6 (4)	Galoli pairs with Dadu'a, not with the Wetar lects
Galoli, Wetar, Tetun	Galoli+Wetar 12 (8); Galoli+Tetun 6 (4); Wetar+Tetun 0 (0)	Galoli belongs with Wetar-Atauro, with a Tetun layer
Helong, Meto, Rote	Meto+Rote 17 (12); Helong+Meto 10 (8); Helong+Rote 3 (2)	Rote-Meto holds; Helong outside it

on the independent lists, but against Tetun only with Tetun Suai: with Tetun Dili, the lingua franca, Lakalei-Idate + Tetun is the larger pair in 84% of combinations, and with the two Tetun lists pooled it wins by one concept (45 to 44).

8 Results: other disputed branches

9 Discussion

What the vocabulary shows. There is no lexical support for Hull's Fabronic, for Timor-Babar or for Eastern Timor, under every coding, with an improved classifier and with lists independent of Grimes and Edwards. This does not refute Timor-Babar, which Grimes and Edwards define by sound changes rather than by shared vocabulary; it means that, if Timor-Babar is a subgroup, its lexicon has since been reshaped by contact. Welaun belongs with Central Timor, and its Tetun appearance sits in borrowable vocabulary, in agreement with Edwards (2019); that contact is the one signal that passes every check. The low-level groups hold, and Galoli pairs with Dadu'a.

Lakalei-Idate remains unresolved, but the balance has moved. Every tree places it beside Mambae, Kemak and Tokodede, now also on lists collected independently of Grimes and Edwards (97%). The two arguments that seemed to favour the eastern placement no longer discriminate: the coded sound changes cost the same on either tree, and the borrowing simulation cannot separate the placements once borrowing runs both ways. The contact readings in the other direction have weakened too, since none of Lakalei-Idate's eastern excesses stands out once unequal rates are modelled. What remains against the Central Timor placement is the qualitative weight Grimes and Edwards give to *p and *ŋ as innovations, and the pull of Tetun Dili on stable vocabulary. The reading that fits most of the evidence is an old, conservative member of a central Timor dialect chain, close to Hull's Ramelaic, which broke away about when Central Timor itself diversified and kept ties to the east.

What would settle it. A specialist's review of the 24 cognate sets behind Lakalei-Idate's stable-vocabulary signal (the packet is in the repository); Grimes and Edwards's 494-item field

lists for Idaté and Lakalei, which are not public; Alcantara's full 800-word Idaté corpus; and a full regular-correspondence analysis of complete wordlists, which these short lists cannot support.

10 Limitations

Apart from two replications with predictions stated in advance, the analyses after the frozen protocol are post hoc; after correction, only Welaun's Tetun contact is statistically secure. The confirmatory predictions failed against Tetun in both replications. The cognates are automatic and have not been checked by a specialist; a consensus of five codings and a validated classifier reduce, but do not remove, that dependence, and the strict consensus favours recent loans. LexiRumah is still a single compilation, although its Idaté, Lakalei and replacement Central Timor lists come from collectors other than Edwards and Grimes. Some sound-change cells for Idaté and Lakalei rest on a handful of tokens. The calibrated triplet test has low power, so not detecting contact is weak evidence against it; the ABC cannot separate the placements; the correspondence test is underpowered. Hull (1998) and Carvalho (2001) could not be obtained and are cited from secondary sources.

Data and code

All data, code, results, the protocol and its addenda are at `https://github.com/DenC-dev/sanak`. The numbered scripts in `src/` rerun every analysis from pinned upstream data. The derived datasets (Grimes and Edwards's tables as data, expert and automatic reflex atlases, sound-change matrices, loan classifications, the Alcantara Idaté list, the retention or innovation status of every cognate set, and the specialist review packet) are in `derived/`. Code is MIT-licensed; data keep their sources' licences.

Use of AI

The code, the analyses and the drafting of this paper were carried out with substantial assistance from Claude (Anthropic), an AI system, under the author's direction. Every analysis, decision and correction is recorded in the public repository so that it can be checked.

References

Albuquerque, Davi Borges de. 2009. Pré-história, história e contatos lingüísticos em Timor Leste. *Domínios de Lingu@gem* 3(2): 75–93.

Alcantara, Maressa Xavier. 2015. *Descrição fonética e fonológica da língua idaté do Timor Leste*. MA thesis, Universidade de São Paulo.

Benjamini, Yoav and Yosef Hochberg. 1995. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society B* 57(1): 289–300.

Dawson, Virginia. 2014. Manatuto survey word lists. PARADISEC collection VD1. In Kaiping, Edwards and Klamer (2019).

Edwards, Owen. 2019. Reintroducing Welaun. *Oceanic Linguistics* 58(1): 31–58.

Edwards, Owen. 2021. *Rote-Meto Comparative Dictionary*. Canberra: ANU Press.

Fogaça, Jessé Silveira. 2015. *Aspectos gramaticais da língua Makasae de Timor-Leste*. PhD thesis, Universidade de Brasília.

Gray, Russell D., Alexei J. Drummond and Simon J. Greenhill. 2009. Language phylogenies reveal expansion pulses and pauses in Pacific settlement. *Science* 323: 479–483.

Greenhill, Simon J., Robert Blust and Russell D. Gray. 2008. The Austronesian Basic Vocabulary Database. *Evolutionary Bioinformatics* 4: 271–283.

Grimes, Charles E. and Owen Edwards. 2026. *The Austronesian languages of eastern Indonesia and Timor-Leste: Unravelling their prehistory and classification*. Berlin: Language Science Press.

Holm, Sture. 1979. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics* 6(2): 65–70.

Hull, Geoffrey. 1998. The basic lexical affinities of Timor's Austronesian languages: a preliminary investigation. *Studies in Languages and Cultures of East Timor* 1: 97–202.

Hull, Geoffrey. 2001. O mapa linguístico de Timor Leste: uma orientação dialectológica. *Studies in Languages and Cultures of East Timor* 4: 1–19.

Kaiping, Gereon A. and Marian Klamer. 2022. The dialect chain of the Timor-Alor-Pantar language family: a new analysis using systematic Bayesian phylogenetics. *Language Dynamics and Change* 12(2): 274–326.

Kaiping, Gereon A., Owen Edwards and Marian Klamer, eds. 2019. *LexiRumah* 3.0. Leiden University Centre for Linguistics.

Kalyan, Siva and Alexandre François. 2018. Freeing the comparative method from the tree model: a framework for historical glottometry. *Senri Ethnological Studies* 98: 59–89.

Klamer, Marian. 2002. Timor-Leste survey word lists. In Kaiping, Edwards and Klamer (2019).

List, Johann-Mattis, Simon J. Greenhill and Russell D. Gray. 2017. The potential of automatic word comparison for historical linguistics. *PLOS ONE* 12(1): e0170046.

Neureiter, Nico et al. 2022. Detecting contact in language trees: a Bayesian phylogenetic model with horizontal transfer. *Humanities and Social Sciences Communications* 9.

Pudlo, Pierre, Jean-Michel Marin, Arnaud Estoup, Jean-Marie Cornuet, Mathieu Gautier and Christian P. Robert. 2016. Reliable ABC model choice via random forests. *Bioinformatics* 32(6): 859–866.

Purwa, I Made et al. 1994. *Struktur bahasa Idate*. Jakarta: Pusat Pembinaan dan Pengembangan Bahasa.

Robinson, Laura C. and Gary Holton. 2012. Internal classification of the Alor-Pantar language family using computational methods applied to the lexicon. *Language Dynamics and Change* 2(2): 123–149.

Tadmor, Uri. 2009. Loanwords in the world's languages: findings and results. In Martin Haspelmath and Uri Tadmor (eds.), *Loanwords in the World's Languages: A Comparative Handbook*, 55–75. Berlin: De Gruyter Mouton.

Wichmann, Søren, Eric W. Holman and Cecil H. Brown, eds. *The ASJP Database*, version 21. Lexibank CLDF release.